**Notes: Deletion Formula for $Y_M(x)$** August 27, 2025

---

Hi everyone,

I'm writing to provide an update on the deletion formula for Y and to ask for some guidance.

Following Tom's suggestion, I've been working on proving the formula by defining the new basis $\mu_F$. The polynomial $\hat{Y}_M(x)$ is defined as

$$\hat{Y}_M(x) = \sum_{F \in \mathcal{L}(M)} \left( x^{\mathrm{crk}(F)} \mu[F, E] \right) \cdot \hat{Q}_{M^F}(x).$$

In the initial steps of the proof, I realized that we might need to redefine $\mu_F$ as

$$\mu_F = \sum_{G \leq F} x^{\mathrm{rk}(G) - 2\mathrm{rk}(F)} \mu[G, F] \cdot G$$

so that the polynomial $\hat{Y}_M(x)$ appears in the expression for $\mu_M$ as

$$\mu_M = \sum_{G \in \mathcal{L}(M)} x^{-\mathrm{rk}(G)} \hat{Y}_{M/G}(x^{-2}) \cdot \zeta^G$$

The issue arises when computing $\Delta(\mu_M)$. It seems that my current approach doesn't satisfy the desired property $\Delta(\mu_M) = \mu_{M \setminus e}$.

I have attached a PDF that contains the detailed LaTeX write-up. It outlines the steps I have taken and the calculations.

It's very likely that I am either misunderstanding a subtle part of the original suggestion or overlooking a crucial detail. Tom, I was hoping that by seeing my specific approach, you might be able to identify where I've gone wrong and provide some guidance.

Thank you for your help.

Best regards,

Nutan

$$F \mapsto x^{\mathrm{rk}(F \setminus e) - \mathrm{rk}(F)} \cdot (F \setminus e)$$